

JOHN PARKER WARD

Conroe, TX 77304 | (832) 779-6235 | johnward326@outlook.com

PROFESSIONAL SUMMARY

Senior Data Engineer with extensive experience designing cloud-native data platforms and real-time pipelines across AWS, Azure, and GCP. Delivered multi-cloud solutions that increased processing throughput by 45% and reduced compute costs by 40% while implementing data mesh and observability frameworks. Skilled in Spark, Databricks, Kafka, dbt, and Airflow to modernize legacy systems and enable enterprise ML workloads. Seeking to apply this expertise to accelerate analytics and drive reliable data solutions as a Senior or Lead Data Engineer.

TECHNICAL SKILLS

- **Programming:** Python, SQL, Scala, Shell, JavaScript, T-SQL
- **Cloud Platforms:** Azure (ADF, Databricks, Synapse, ADLS), AWS (Glue, Redshift, EMR, Lambda, S3), GCP (BigQuery, Dataflow, Composer)
- **Data Engineering:** Spark, PySpark, Databricks, Kafka, Kinesis, Flink, Airflow, dbt, Delta Lake, Iceberg, DLT, SQL Warehouse
- **Databases & Warehousing:** Snowflake, Redshift, Synapse, PostgreSQL, DynamoDB, MySQL, BigQuery, SQL Server
- **DevOps & Infrastructure:** Terraform, CloudFormation, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, IaC
- **Data Quality & Observability:** Great Expectations, Monte Carlo, Soda, OpenLineage, Prometheus, Grafana, ELK
- **Other:** REST APIs, GraphQL, Parquet, Avro, JSON, MLflow, Feature Stores, Data Modeling, SCD Type-2

PROFESSIONAL EXPERIENCE

CapTech Consulting

Dec 2022 - Present

Senior Data Engineer

Conroe, TX

- Architected multi-cloud data platforms using Databricks, Delta Lake, Snowflake, and Synapse, boosting enterprise processing throughput by 45%.
- Engineered high-volume real-time and batch pipelines using Spark, Kafka, Kinesis, and Airflow, delivering analytics, ML scoring, and operational intelligence that improved decision-making speed for stakeholders
- Designed domain-oriented data mesh architectures and reusable ingestion frameworks, accelerating new pipeline delivery by 40%.
- Established enterprise observability standards with Great Expectations, Monte Carlo, and OpenLineage, reducing data quality incidents by 35%.
- Automated infrastructure provisioning with Terraform and Kubernetes, enabling consistent, repeatable deployments across client environments.
- Defined governance, lineage, and access-control frameworks using Unity Catalog and schema registry, enabling consistent cross-domain data standards and reducing unauthorized data-access incidents
- Built ML feature pipelines and automated model-scoring workflows on Databricks, accelerating delivery of predictive analytics and improving model-scoring reliability
- Designed event-driven streaming architectures for IoT data using Kafka, Kinesis, and Flink, reducing data latency to under a minute and supporting real-time analytics
- Developed dbt transformation layers with automated testing, CI/CD integration, and version-controlled deployments, ensuring data quality and speeding up release cycles
- Delivered data platform roadmaps and architecture blueprints for Fortune 500 clients, guiding Azure-based implementations that lowered infrastructure costs and aligned the platform with business objectives
- Mentored engineering teams on applying Azure Data Factory and dbt design patterns, performance-tuning SQL queries, and building scalable pipelines, which improved processing speed and code consistency across projects
- Led research on Iceberg, Delta Sharing, and Lakehouse Federation using Azure Synapse and Databricks, delivering proof-of-concepts that enabled clients to modernize their data platforms and reduce data latency

Target Corporation

Apr 2018 - Nov 2022

Senior Data Engineer

Minneapolis, MN

- Engineered large-scale ETL/ELT pipelines using Spark, Airflow, and Databricks to process billions of retail transactions and product events, enabling timely reporting for business stakeholders
- Led migration from Teradata/Oracle to Snowflake and AWS S3, cutting compute costs by 40% and improving query performance.
- Built real-time ingestion pipelines with Kafka and Spark Streaming for inventory, pricing, and supply chain analytics, delivering near-real-time insights that supported faster decision-making
- Developed dbt models to standardize analytical datasets, enforce business logic, and improve data consistency across domains.
- Designed and maintained 500+ Airflow DAGs with SLA tracking, automated alerting, and dependency-aware orchestration, which reduced workflow failures and ensured data pipelines met delivery targets
- Implemented data validation, monitoring, and lineage frameworks using Great Expectations and Monte Carlo, enhancing data quality and auditability for governance
- Partnered with Data Science teams to build ML-ready datasets in BigQuery using SQL and Parquet, accelerating forecasting, personalization, and demand-modeling projects

- Automated infrastructure provisioning using Terraform and CloudFormation, enabling consistent multi-environment deployments.
- Optimized Spark jobs through partitioning, caching, and I/O tuning, improving pipeline performance by 40%.
- Built Power BI dashboards to surface KPIs for merchandising, supply chain, and operational leadership.
- Developed metadata-driven ingestion frameworks with Great Expectations, Docker, and GCP (BigQuery, Dataflow, Composer) that supported schema evolution, enabled dynamic transformations, and automated onboarding, reducing pipeline setup time and improving data quality

Fivetran

Jun 2014 - Mar 2018

Data Engineer

Oakland, CA

- Built and maintained high-throughput ETL pipelines with PySpark, Python, and SQL on AWS Redshift, enabling reliable connector ingestion and automated ELT workflows for downstream analytics
- Developed metadata-driven ingestion frameworks enabling dynamic schema detection, incremental loading, and parameterized transformations.
- Designed star and snowflake schema models for internal analytics and customer-facing reporting, improving query efficiency.
- Built automated ingestion workflows from APIs, SaaS platforms, and RDBMS sources into AWS S3 using CloudFormation and Kubernetes, adding robust error handling that lowered data-load failure rates
- Created Airflow DAGs to schedule jobs, manage dependencies, and trigger automated recovery across distributed systems, maintaining high availability of data pipelines
- Tuned SQL and Spark workloads to reduce processing time by 25% and improve connector reliability.
- Collaborated with analysts to streamline reporting pipelines using Looker and Power BI, delivering reports faster and reducing data discrepancies
- Supported CI/CD pipelines with Jenkins and Git-based workflows, automating testing and deployment, which shortened release cycles and lowered deployment errors
- Developed Python-based validation and reconciliation tools to ensure accuracy across large-scale datasets, decreasing data-quality issues and reducing manual review effort
- Built internal tooling to improve connector performance, handle schema evolution, and monitor ingestion, resulting in more stable pipelines and quicker issue detection
- Migrated legacy ingestion logic to a modular, scalable Python framework, improving maintainability and throughput.
- Enhanced connector performance by optimizing API pagination, batching, and incremental sync logic.

EDUCATION

University of Texas at Austin

Aug 2010 - May 2014

Bachelor of Science, Computer Science